

Assignment-01

Computer Networks – Network Related for Windows 10 / 11 Commands

IP & Network Configuration

- `ipconfig`
Displays basic TCP/IP configuration.
- `ipconfig /all`
Displays detailed IP configuration for all adapters.
- `ipconfig /release`
Releases the current DHCP lease.
- `ipconfig /renew`
Renews the DHCP lease.
- `ipconfig /flushdns`
Clears the DNS resolver cache.
- `ipconfig /displaydns`
Displays cached DNS records.

Connectivity Testing

- `ping www.google.com`
Tests connectivity to a host.
- `ping -t www.google.com`
Sends continuous ICMP echo requests.
- `ping -l 1024 www.google.com`
Sends ping with a specified packet size.
- `ping -a 192.168.1.1`
Resolves IP address to hostname.
- `tracert www.google.com`
Traces the route packets take to a destination.
- `pathping www.google.com`
Combines ping and traceroute with packet-loss statistics.

Network Statistics & Connections

- `netstat`
Displays active connections.
- `netstat -an`
Displays all connections and listening ports numerically.
- `netstat -r`
Displays routing table.

DNS & Name Resolution

- `nslookup www.google.com`
Queries DNS servers for domain or IP information.

ARP & Routing

- `arp -a`
Displays ARP cache (IP–MAC mappings).
- `route print`
Displays the routing table.
- `route add 192.168.2.0 mask 255.255.255.0 192.168.1.1`
Adds a static route.

Network Interface & Configuration (netsh)

- `netsh interface ip show config`
Displays IP configuration for interfaces.
- `netsh interface ipv4 show dns`
Displays DNS configuration.
- `netsh interface ip set address name="Ethernet" static 192.168.1.100 255.255.255.0 192.168.1.1`
Sets a static IP address.
- `netsh interface ip set dns name="Ethernet" static 8.8.8.8`
Sets a static DNS server.

Wireless Networking

- `netsh wlan show interfaces`
Displays wireless interface details.
- `netsh wlan show profiles`
Displays saved Wi-Fi profiles.
- `netsh wlan connect name="NetworkName"`
Connects to a wireless network.
- `netsh wlan disconnect`
Disconnects from the wireless network.

Firewall Management

- `netsh advfirewall show allprofiles`
Displays firewall status.
- `netsh advfirewall set allprofiles state off`
Turns off firewall (Administrator required).
- `netsh advfirewall set allprofiles state on`
Turns on firewall.

File Sharing & Network Resources

- `net use Z: \\servername\sharename`
Maps a network drive using hostname.
- `net use Z: \\192.168.1.10\sharename`
Maps a network drive using IP address.
- `net share sharename=C:\folderpath`
Shares a local folder.
- `net view \\servername`
Displays shared resources on a server.

Network Services & Statistics

- `net start servicename`
Starts a network service.
- `net stop servicename`
Stops a network service.
- `net statistics workstation`
Displays workstation statistics.
- `net statistics server`
Displays server statistics.
- `net config server`
Displays server service configuration.
- `nbtstat -a hostname`
Displays NetBIOS over TCP/IP statistics.

PowerShell Network Commands

Configuration & Status

- `Get-NetAdapter`
- `Get-NetIPAddress`
- `Get-NetIPConfiguration`
- `Get-NetRoute`

Connectivity & DNS

- `Test-Connection www.google.com`
- `Resolve-DnsName www.google.com`
- `Get-DnsClientCache`

Firewall & Interface Control

- `Get-NetFirewallProfile`

- Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False
- Enable-NetAdapter -Name "Ethernet"
- Disable-NetAdapter -Name "Ethernet"

Advanced Configuration

- New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 192.168.1.100 -PrefixLength 24 -DefaultGateway 192.168.1.1
- Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses 8.8.8.8
- Get-NetTCPConnection

Help & Documentation

- command /?
- Get-Help Get-NetIPAddress